UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

a) Describe ONE (1) major difference between a tightly coupled multiprocessor system and a loosely coupled multiprocessor system.

b) State the THREE (3) types of cluster architecture.

QUESTION A2 **(5 marks)**

Convert 100.5 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

QUESTION A3 **(5 marks)**

List the name of the inventor of the following devices:

a) Mechanical Calculating Machine

b) Difference Machine

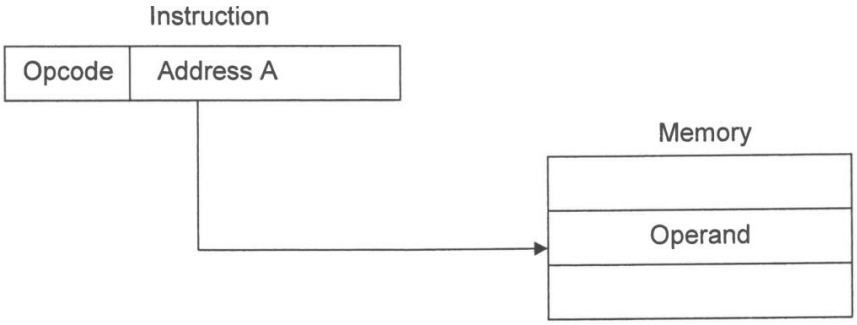
c) Analytical Machine

d) IAS Machine

e) Integrated circuit

QUESTION A4 **(5 marks)**

Given the following Figure 2.



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a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

QUESTION A5 **(10 marks)**

a) Why must the CPU be faster than all the other inpuuoutput devices?

b) How do the I/O controllers facilitate communication between the CPU and I/O devices?

c) Discuss TWO (2) types of I/O modules.

d) When would an I/O controller of a specific device be installed?

QUESTION A6 **(10 marks)**

Answer both parts. Show your complete workflow.

a) Convert 159910 to hexadecimal.

b) Convert FD916 to decimal.

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b) Convert FD916 to decimal.

QUESTION A7 **(10 marks)**

a) Discuss THREE (3) types of interfaces to extemal devices.

b) Classify the following input/output devices to their respective 1/0 module:

i. Network interface card

ii. Scanner Touch Screen

i. Register

QUESTION A8 **(10 marks)**

System interconnection is a mechanism that provides communication among computer components.

a) What is the size of a bus if the word size is 64 bit?

b) List the TWO (2) types of bus.

c) Illustrate the location of a bus between CPU and memory.

d) Differentiate between system interconnection and CPU interconnection.

e) Discuss ONE (1) drawback each for the different types of bus timing.

QUESTION A9 **(10 marks)**

a) Discuss the main function of cache memory.

b) How can the data stored in the cache and main memory be synchronized.

c) Contrast between SRAM and DRAM based on the following characteristics: Type of storage Availability of charges leak price Example of location of memory usage

QUESTION A10 **(10 marks)**

a) Draw a complete machine cycle by giving the exact location of each phase.

b) Briefly, describe each phase in the machine cycle.

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

$$y = A - \frac{2C}{B + A^2} * D$$

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three- address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

One-address	Two-address	Three-address
LOADp	MOV (p,q)	ADD (p.q.r)
STORp	ADD (p,q)	SUB (p,q,r)
ADDp	SUB (p,q)	MUL (p.q.r)
SUBp	MUL (p,q)	DIV(p,q,r)
MULp	DIV(p,q)	
DIVp		